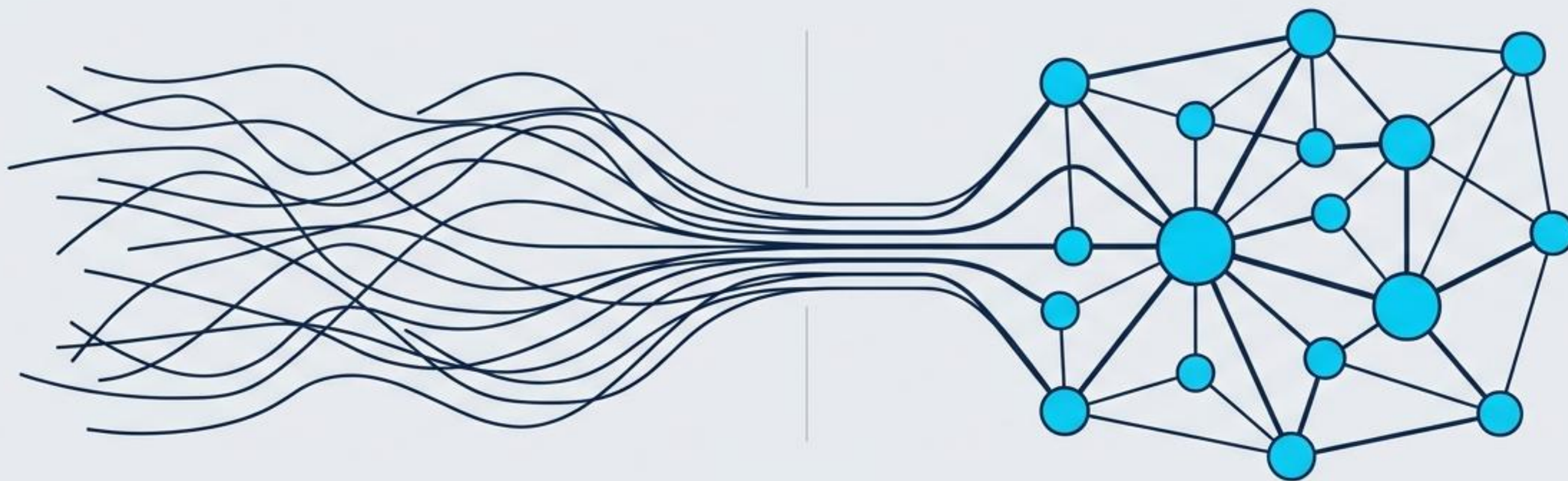


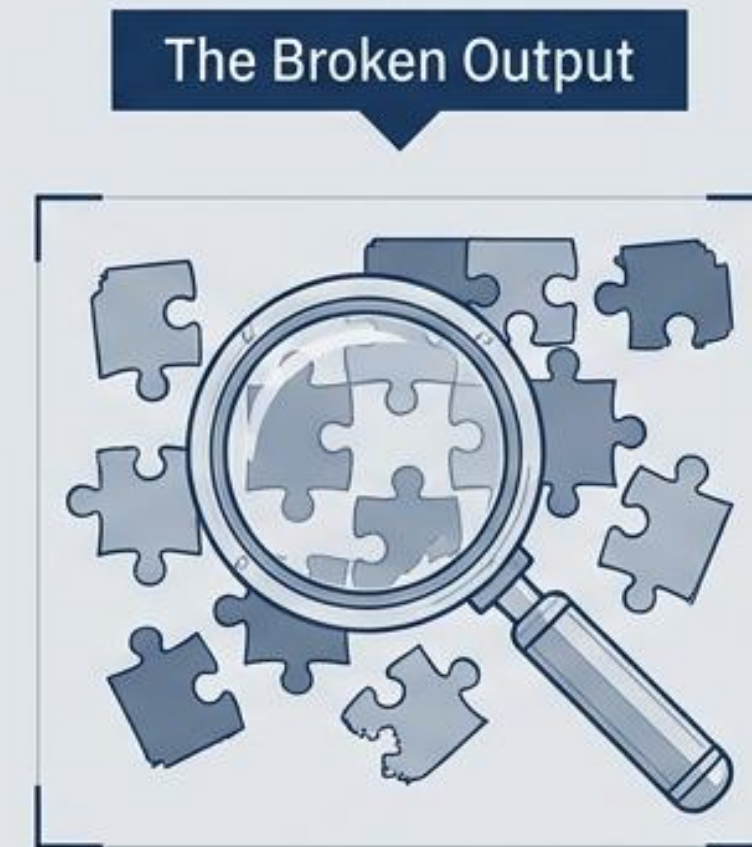
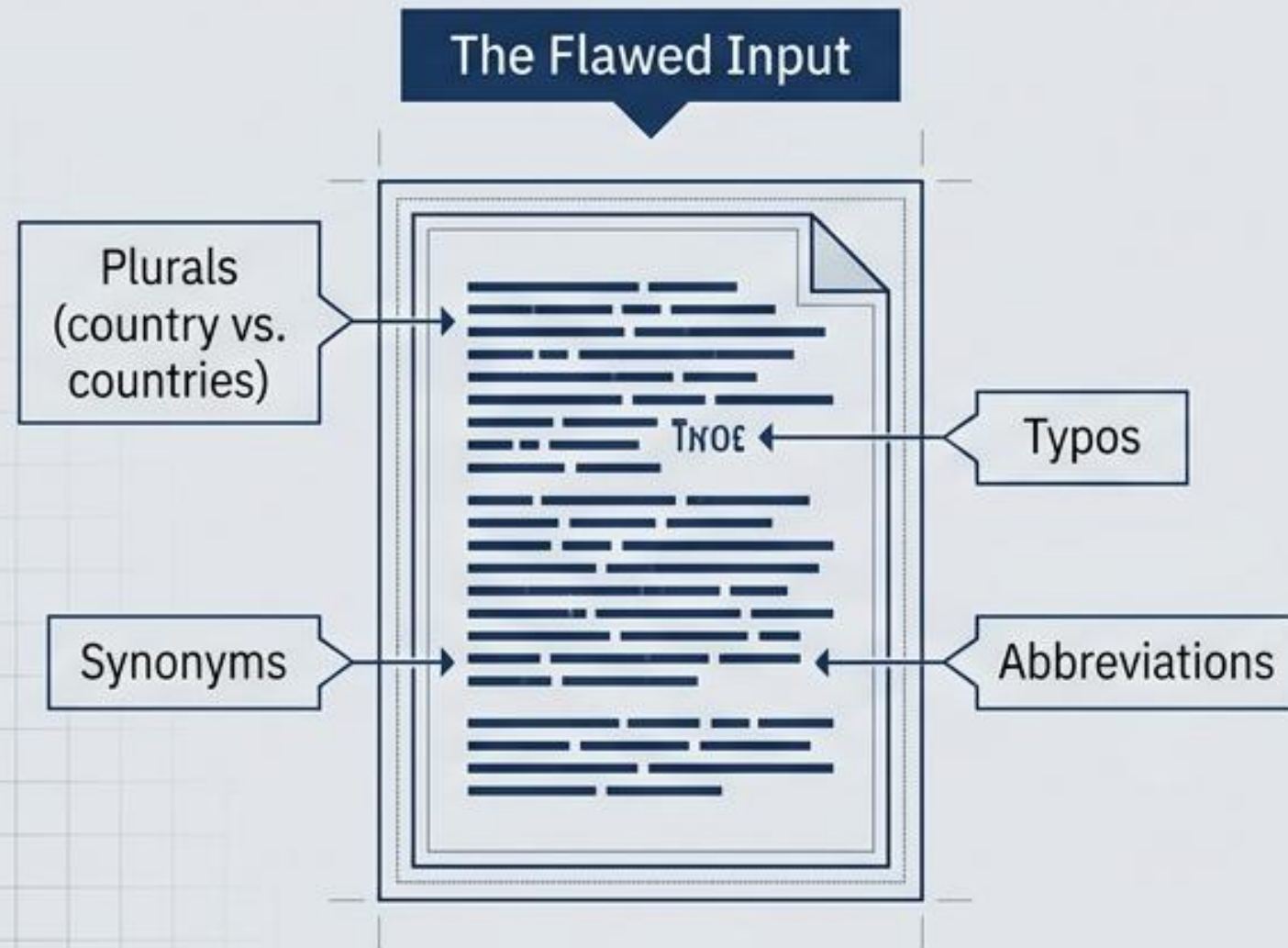
Semantic Search and Similarity



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The Unstructured Data Dilemma

A massive proportion of the world's data exists as unstructured natural language. Legacy search engines rely on simple keyword indexing and word counts to parse this data, leading to inaccurate, incomplete results.



Searches for exact character strings rather than underlying meaning.

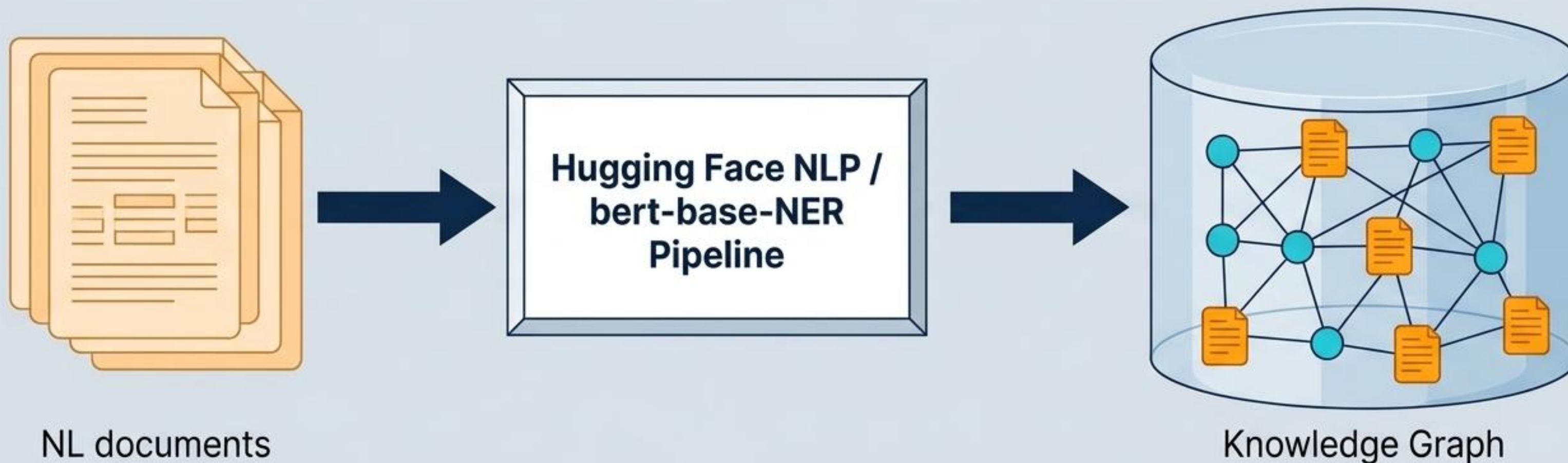
Natural language is ambiguous and contextual. Searching for raw text strings is fundamentally inaccurate and inefficient.

Search Paradigms: A Maturation Matrix

Feature	Stage 1: Legacy Indexing (Strings)	Stage 2: Entity Search (NER)	Stage 3: Semantic Search (Things)
Core Unit	Words / Characters	● Extracted Entities	● Organizing Principles (Ontologies)
Handling Synonyms	Fails (ignores plurals and variations)	● Resolves basic variations	● Mastered via shared concept schemes
Handling Ambiguity	Fails (treats identical words the same)	● Resolves context (e.g., Person vs. City)	Fully disambiguated via domain knowledge
Query Type	Keyword Matching	● Entity Matching	● Concept Traversal
Output	Document list sorted by term frequency	● Documents linked by explicit objects 	 Explainable, multi-dimensional answers

Stage 1: Unlocking Hidden Structures

To move past strings, we apply Natural Language Processing (NLP) —specifically Named-Entity Recognition (NER) —to extract unambiguous entities hidden within human language.

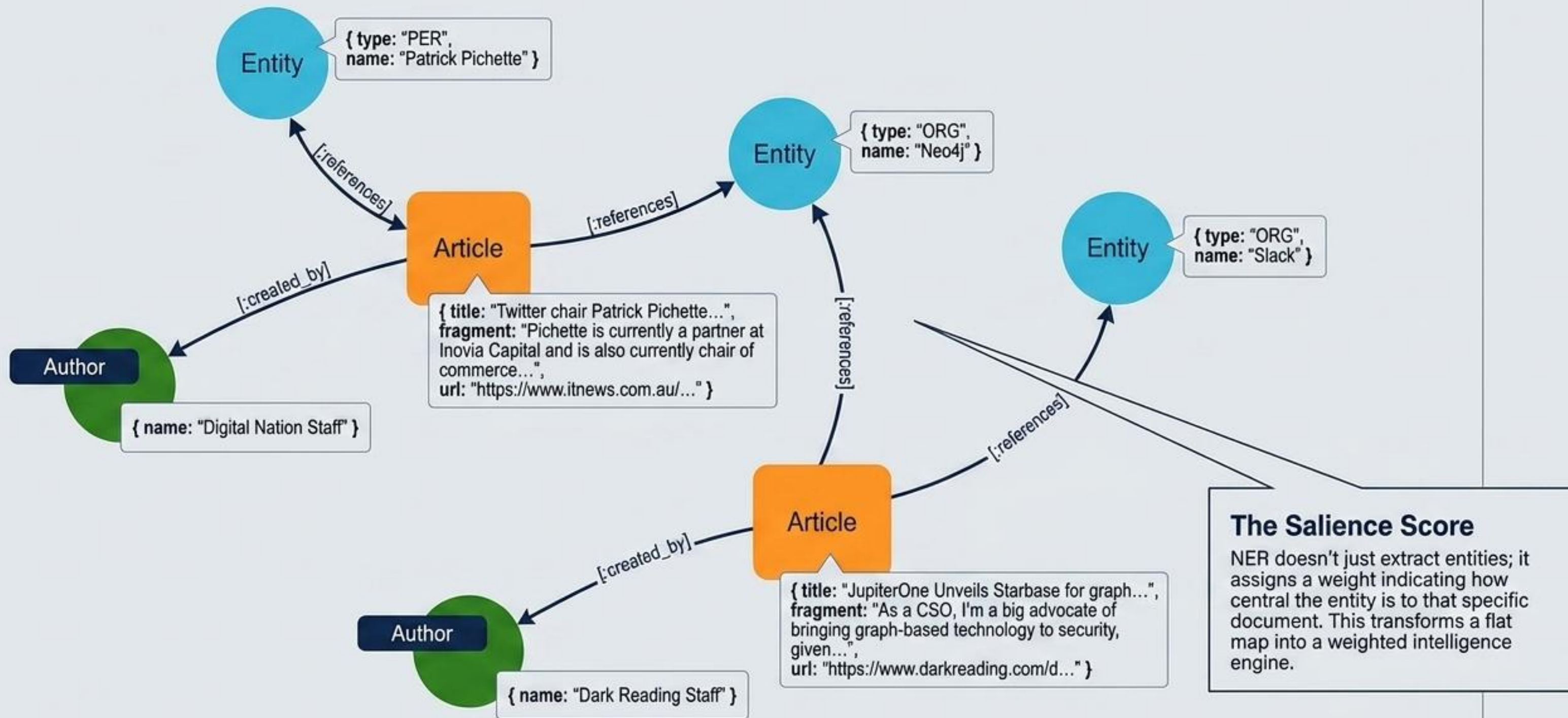


NER Context Example:

The New York ^[ORG] Times is headquartered in New York ^[LOC] City.

NER understands context where string-matching fails.

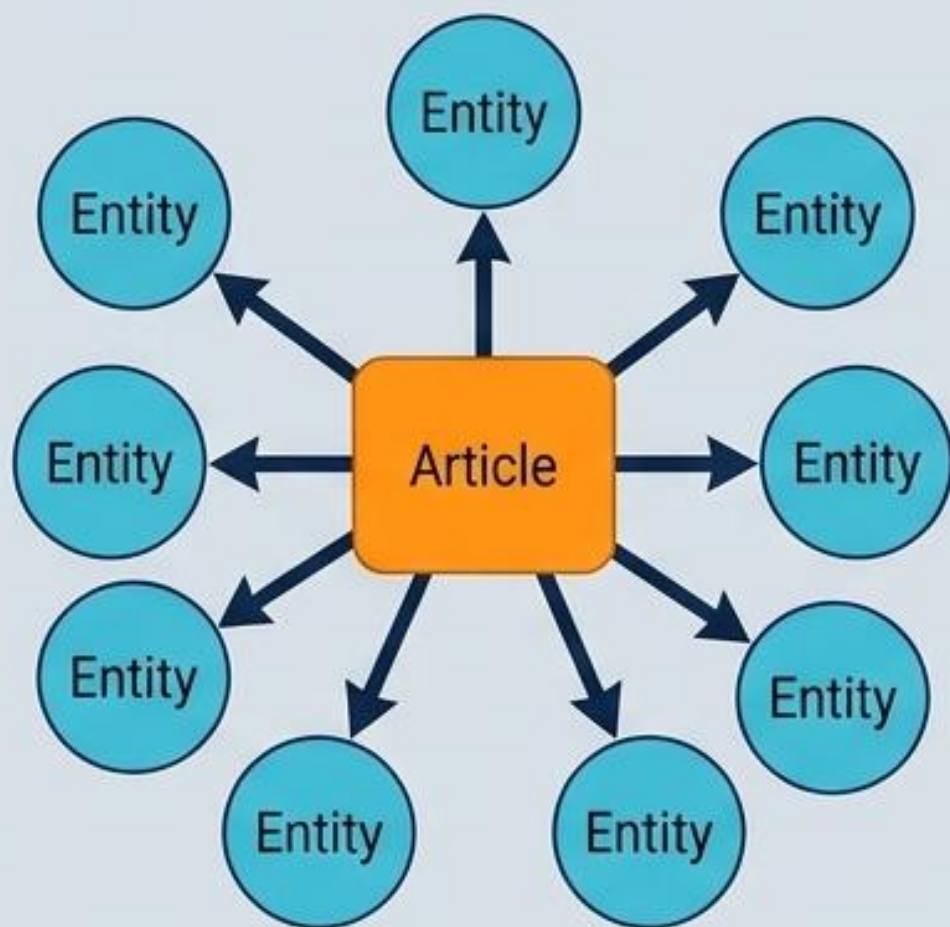
Architecting the Knowledge Graph



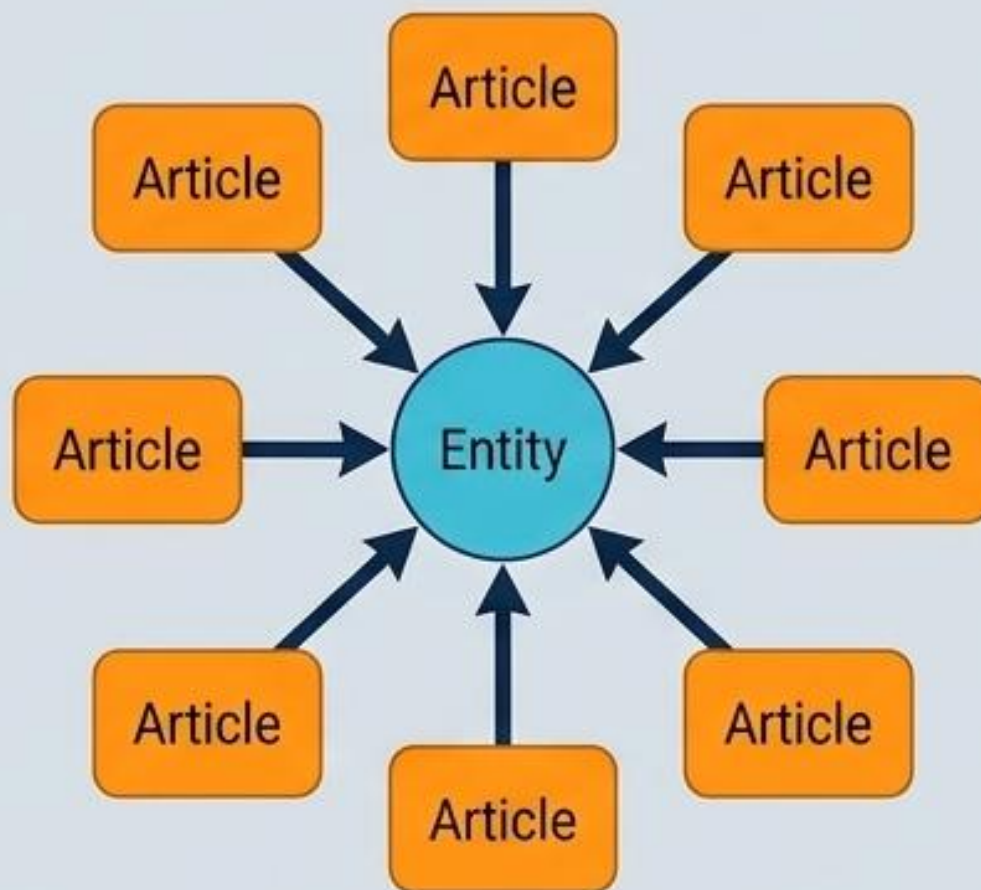
Beyond Word Counts: Graph-Native Analysis

Questions the Graph Can Answer

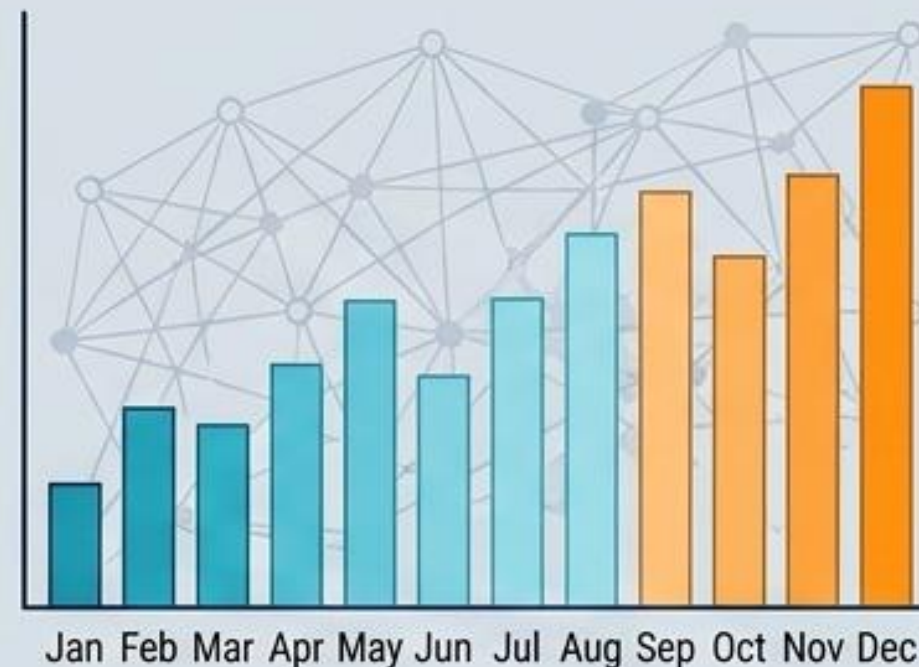
What is this document about?



Which documents mention this entity?



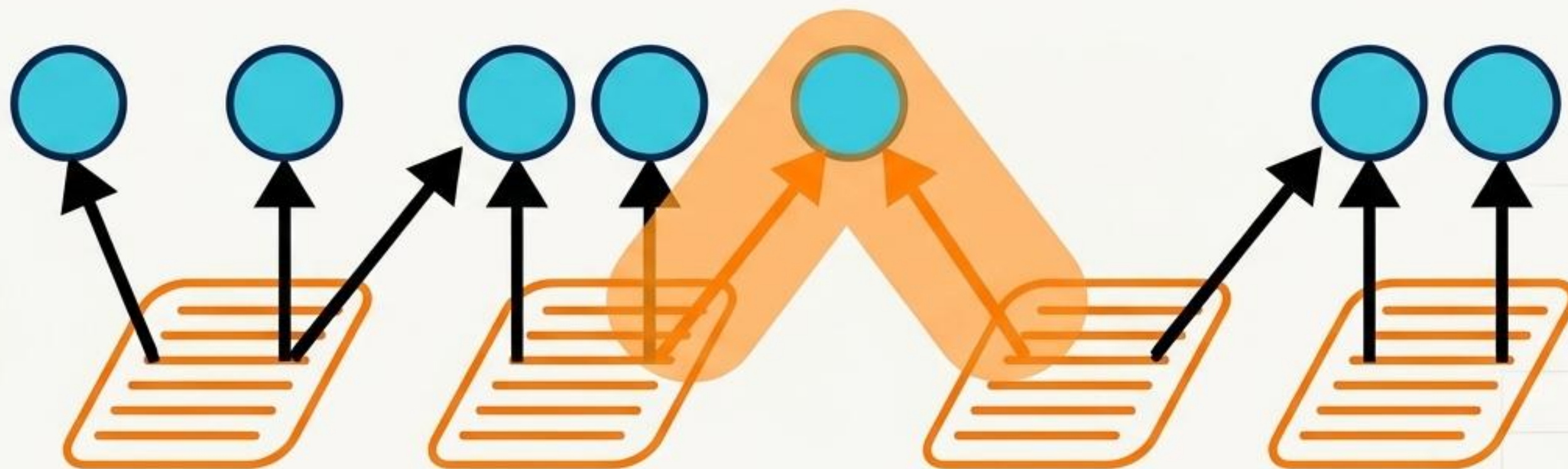
How is this topic trending?



Tracks a concept's true popularity over time, completely independent of specific keyword frequency.

Stage 2: Relational Document Similarity

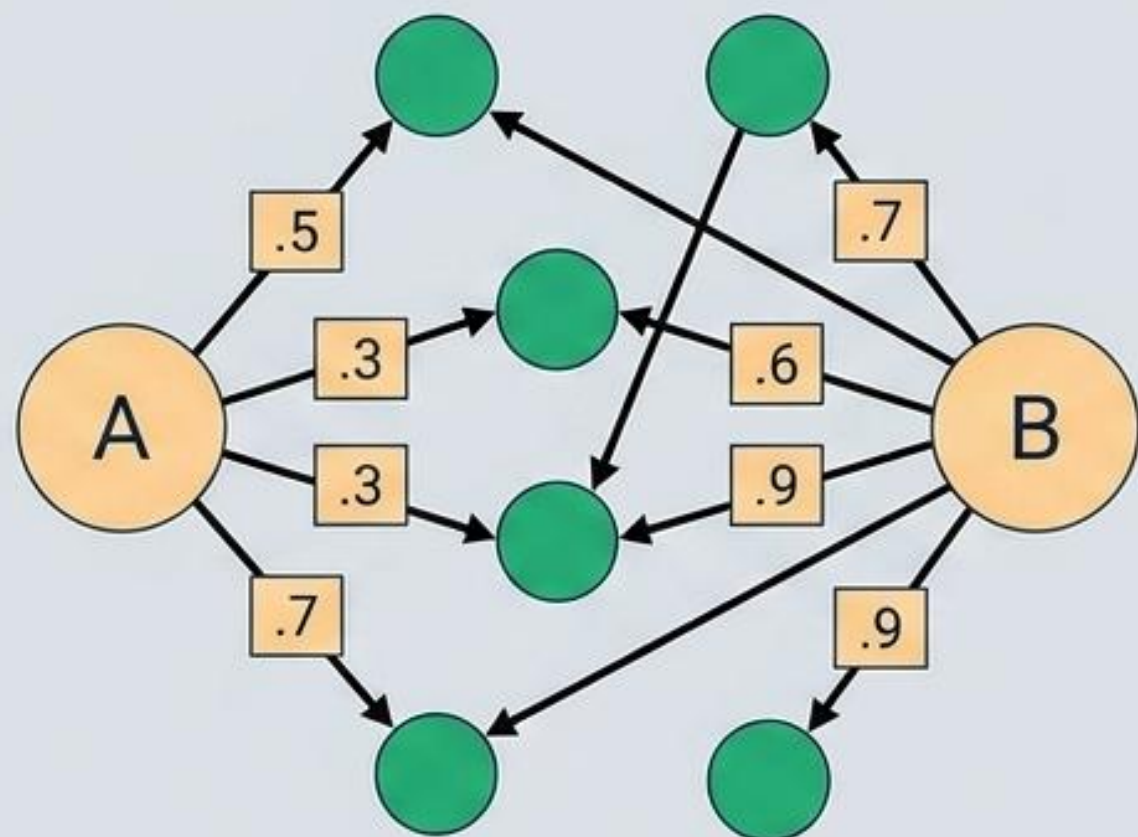
Articles naturally relate to one another through the entities they share. By navigating these connections, the graph powers highly precise, explainable content recommendations ("You may also like X").



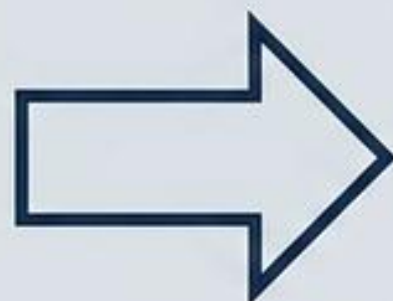
The overlap of shared entities dictates similarity.
The graph explains *why* two documents are related.

The Math of Similarity

The Bipartite Graph



Similarity Metric = $\text{sum}(r1.\text{salience} * r2.\text{salience})$



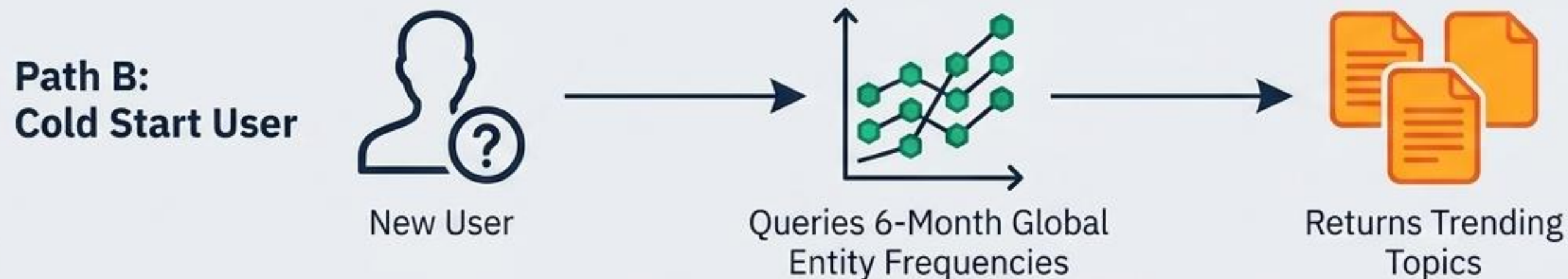
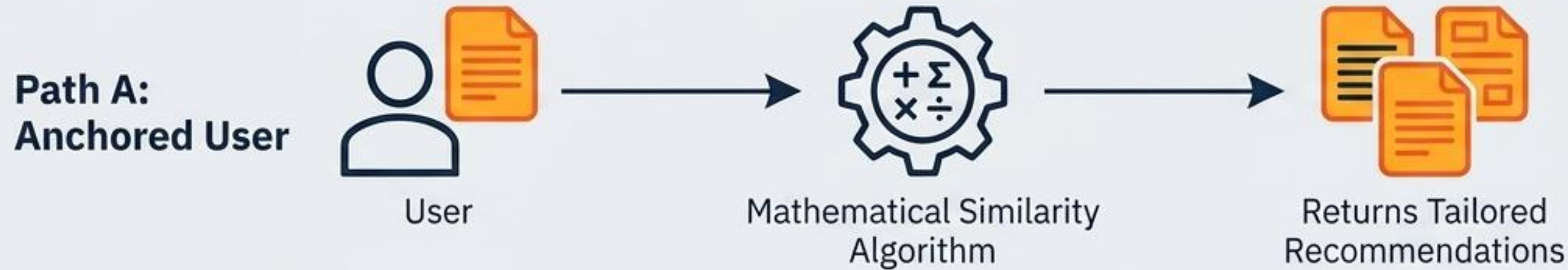
The Monopartite Graph: Similar Articles



The system multiplies the salience weights of shared entities to compute a direct, material similarity score between any two articles.

Overcoming the “Cold Start”

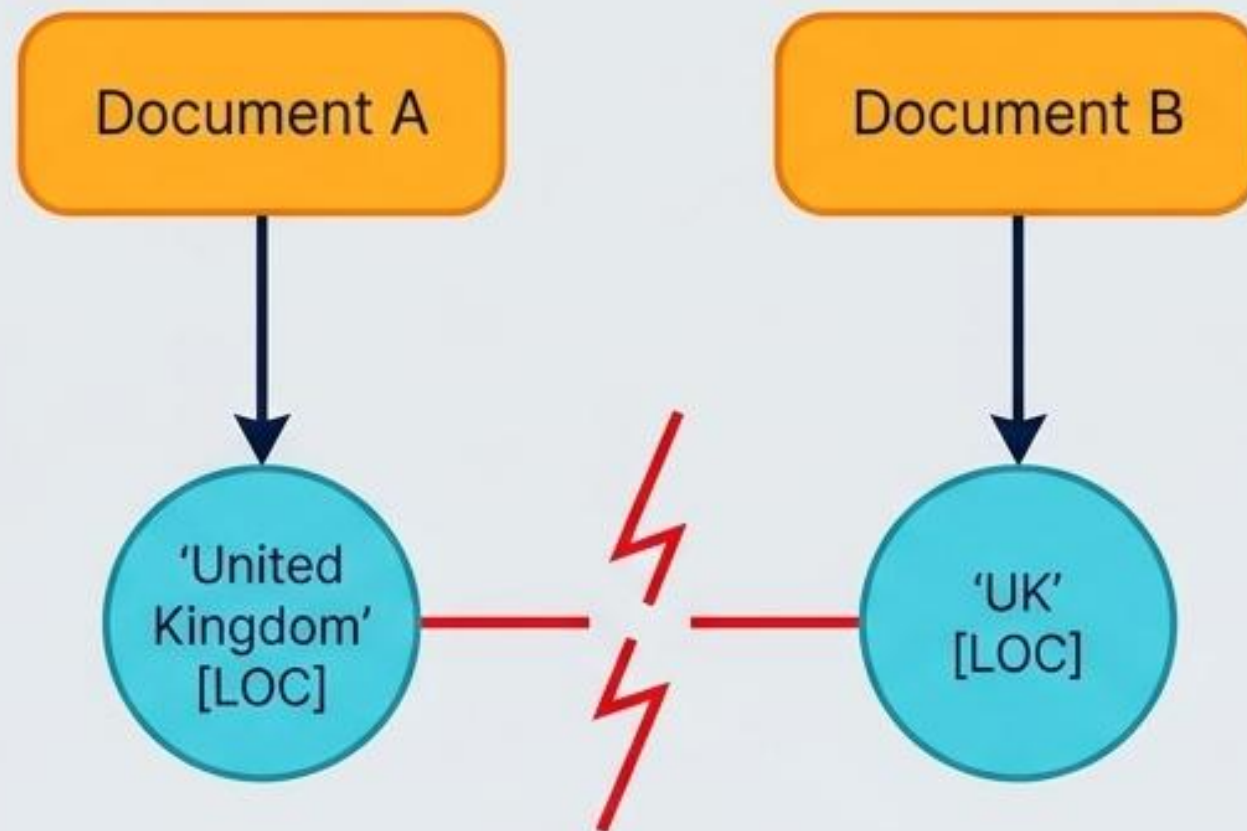
Document similarity requires an anchor (a currently viewed article). What happens when a new user visits the homepage with zero context?



Takeaway: The graph structure allows systems to seamlessly toggle between personalized mathematical similarity and global trend analysis to guarantee relevant content.

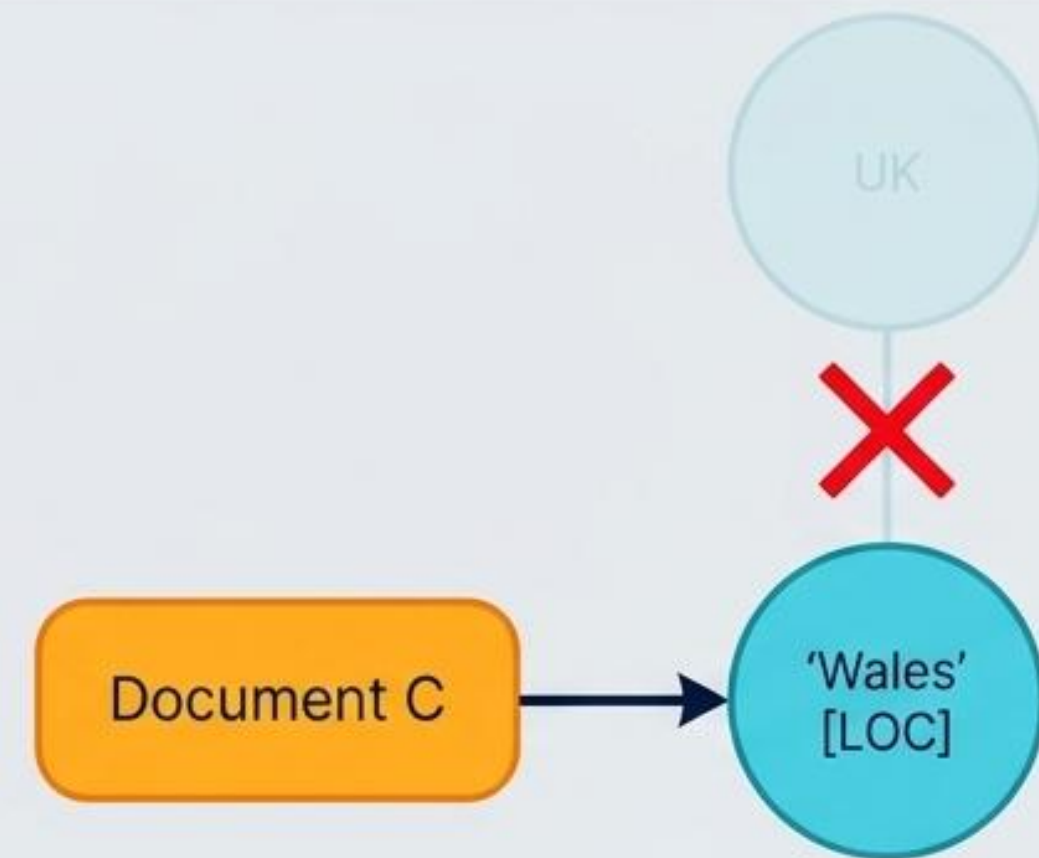
The Disambiguation Hurdle

Scenario 1: Lack of Disambiguation



NER tags both locations but strands them as isolated nodes.

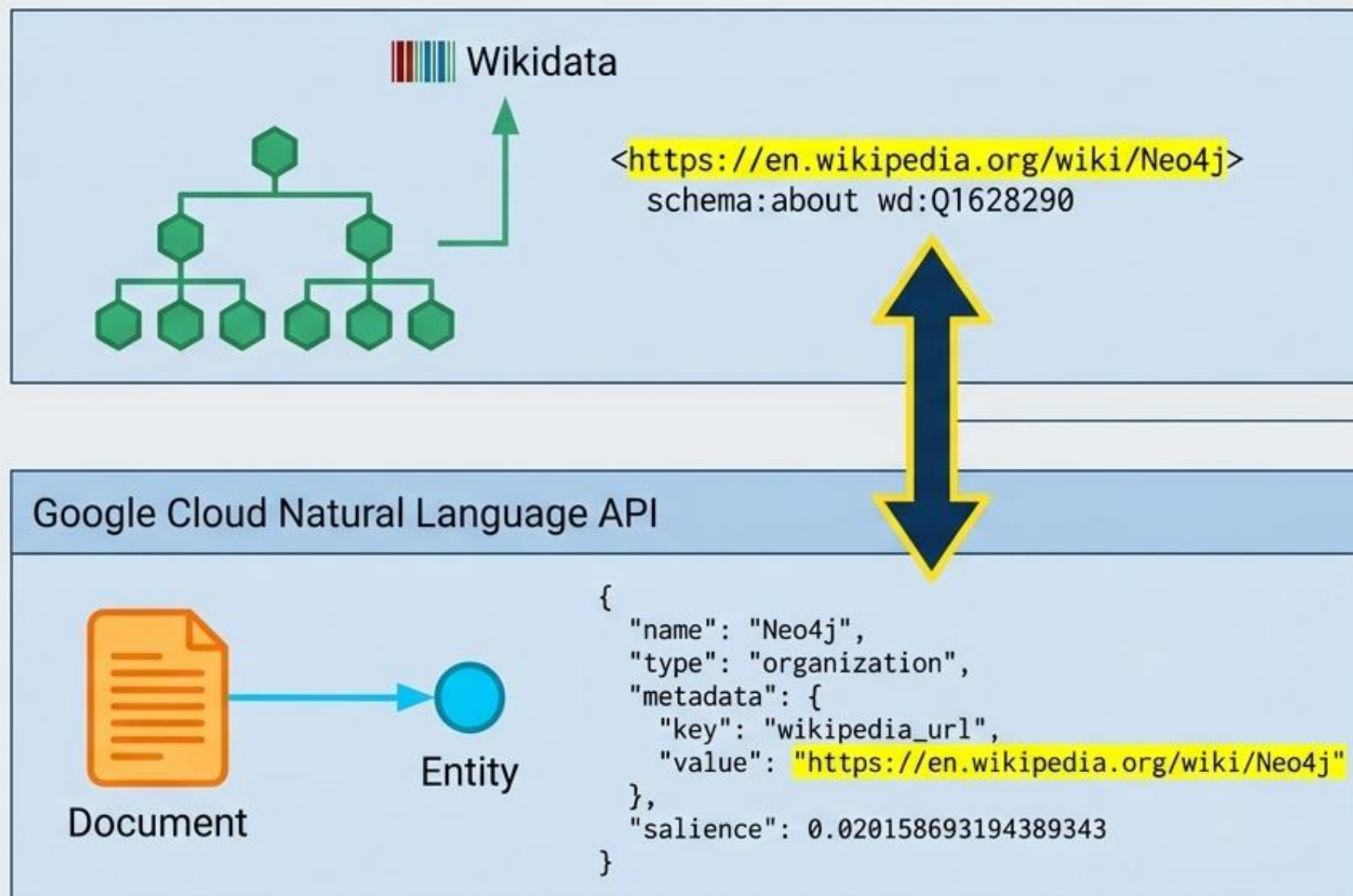
Scenario 2: Lack of Domain Knowledge



The system doesn't know Wales has a [PART_OF] relationship to the UK, hiding relevant content.

To achieve true semantic search, we must map extracted entities against a universal organizing principle.

Stage 3: Entity Linking via Organizing Principles

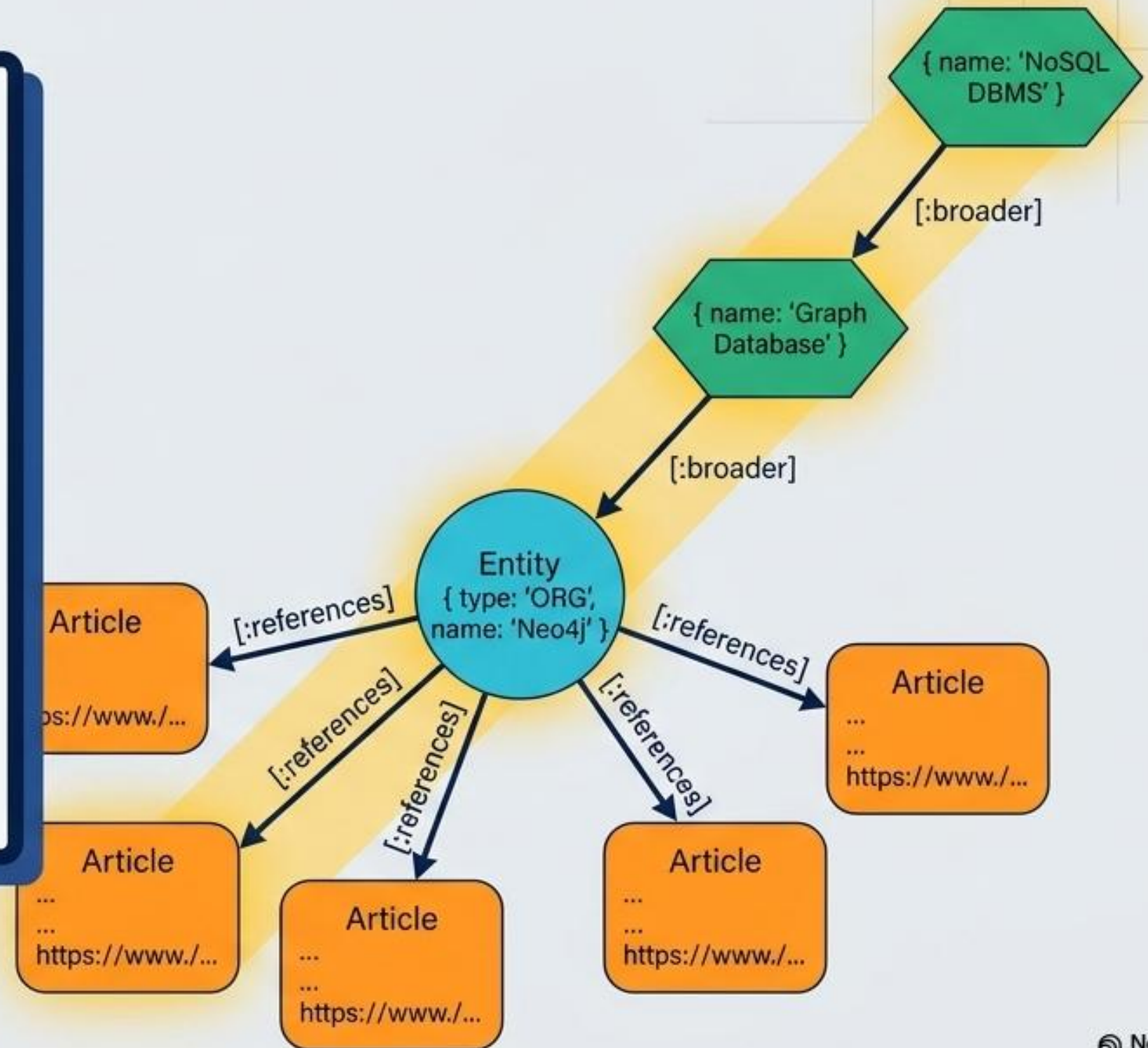


Entities are no longer just guessed strings; they are **definitively locked to unique, globally recognized concepts.**

The Semantic Search Mechanism

The Payoff: A user searches for "NoSQL DBMS". The system successfully returns highly relevant articles about Neo4j, **even if the string 'NoSQL' never appears in the text.**

This is the definition of "Things, not strings."

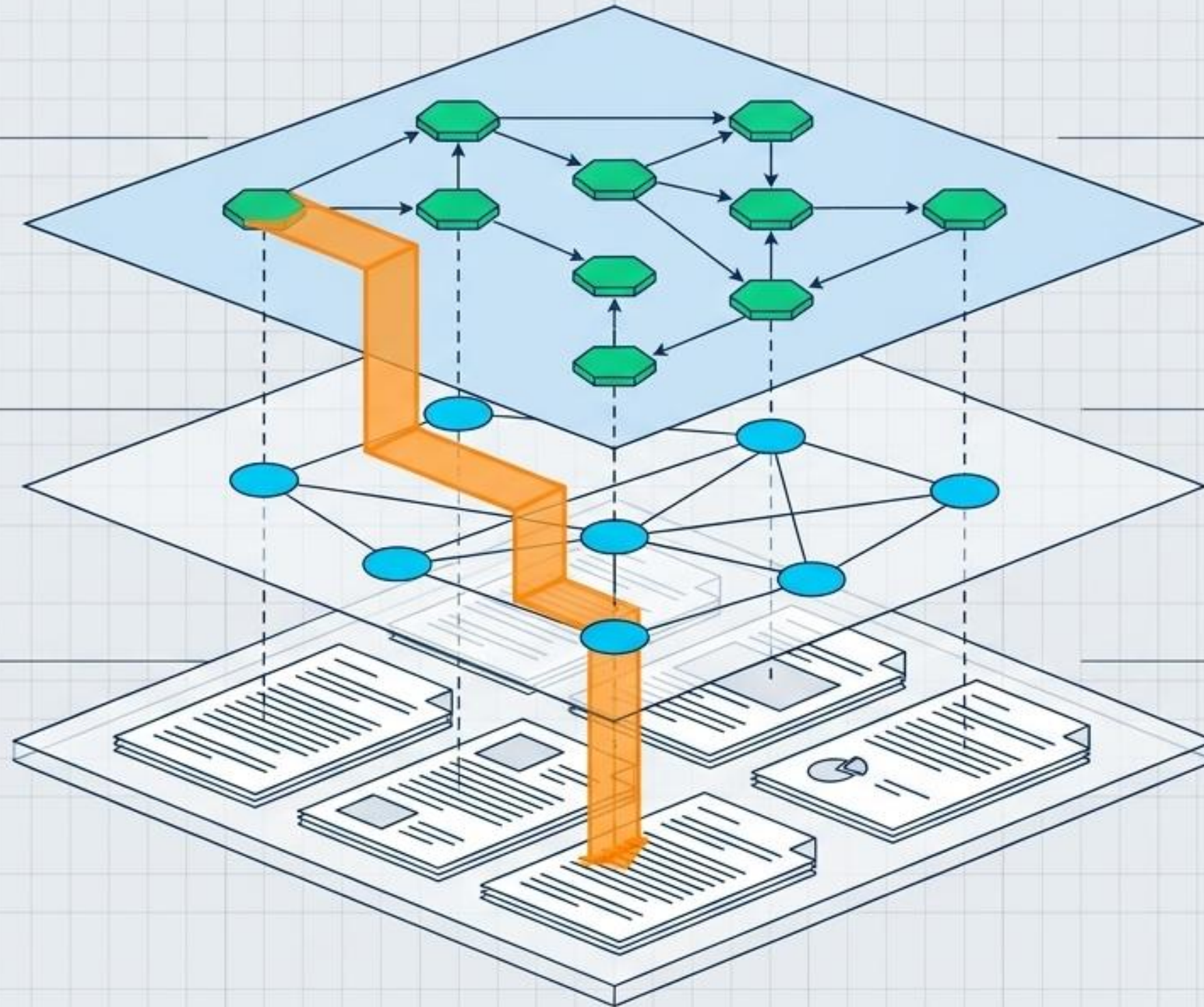


Synthesis: The Three-Tiered Intelligence Architecture

Knowledge Layer:
Organizing Principle

Entity Layer:
Annotation

Text Layer:
Documents



Real-World Impact: NASA Lessons Learned

A case study on NASA's Lessons Learned Information System (LLIS), containing knowledge spanning from the Apollo missions to today.

The Problem

20 million siloed documents. Keyword searches took too long and returned overwhelming, disconnected results (1,100+ documents per query).

The Solution

Transforming LLIS into a Semantic Knowledge Graph. Extracting entities, establishing categories, and correlating topics to uncover preventative trends.

The ROI

"This has saved us at least a year and over \$2M in research and development towards our Mission to Mars planning." — NASA Engineer

